

Chris Cundy

Experience

- June 2024– **Research Scientist**, *FAR.AI*, Berkeley, California, USA
 Characterising and mitigating catastrophic risks from frontier AI systems. My main responsibilities are:
- Leading research projects and determining the direction for teams of researchers and engineers.
 - Designing and implementing training and evaluation schemes with state-of-the-art models, conducting empirical studies, and communicating results through papers and presentations.
 - As a research team lead, ensuring that research aims are consistent with FAR's mission and that my direct reports are supported in their output and professional development.
- June 2022– **Research Scientist Intern**, *Technical AI Safety Team*, DeepMind, London, UK
- September 2022 Investigating robust and reliable machine learning in theory and at scale
- Investigated susceptibility of autoregressive models to 'delusions', where unobserved latent variables lead to incorrect probabilistic judgments.
 - Developed a theoretical model for delusions; investigated delusions at scale by analysing performance of DeepMind's Gato (a large generalist, multi-task autoregressive model) on custom environments.
- October 2017– **Visiting Scholar**, *Future of Humanity Institute*, University of Oxford, Oxford, UK
- January 2018 Developing algorithms to predict human judgments. Supervised by Owain Evans and Andreas Stuhlmüller
- Designed algorithms to collate quick, noisy human judgments to predict the answer to complicated tasks which would typically require deliberation.
- June– **Visiting Scholar**, *Centre for Human-Compatible AI*, University of California, Berkeley, US
- September 2017 Supervised by Daniel Filan & Stuart Russell. I analysed inverse reinforcement learning in the hierarchical options setting. I formalized the problem, performed experiments on data and presented at an ICML workshop.

Education

- 2018-2024 **PhD - Computer Science**, *Stanford University*, Stanford, California, USA
- Advised by Stefano Ermon.
 - Investigating topics in inverse reinforcement learning, sequence modelling and variational inference.
 - Thesis: *Beyond Maximum Likelihood: Distribution-Aware Machine Learning*.
- 2016-2017 **MEng - Computer Science**, *University of Cambridge*, Cambridge, UK
- Grade: Distinction. Modules include: Data Science, Probabilistic Machine Learning, Network Analytics.
 - Supervised by Carl E. Rasmussen.
- 2013-2016 **BA - Natural Sciences (Physics)**, *University of Cambridge*, Cambridge, UK
- Grade: 1st. Modules: Physics, Maths, Chemistry, Computer Science.

Selected Publications

- 2026 **The Obfuscation Atlas: Mapping Where Honesty Emerges in RLVR with Deception Probes**,
Mohammad Taufeeque, Stefan Heimersheim, Adam Gleave, Chris Cundy, ICML 2026 (Outstanding paper honorable mention, top seven (0.1%) among accepted papers)
- 2025 **Auditing Games for Sandbagging**,
Jordan Taylor, Sid Black, Dillon Bowen, Thomas Read, Satvik Golechha, Alex Zelenka-Martin, Oliver Makins, Connor Kissane, Kola Ayonrinde, Jacob Merizian, Samuel Marks, Chris Cundy, Joseph Bloom, Technical Report

- 2025 **Preference Learning with Lie Detectors can Induce Honesty or Evasion**,
Chris Cundy, *Adam Gleave*, NeurIPS 2025
- 2024 **SequenceMatch: Imitation Learning for Autoregressive Sequence Modelling with Backtracking**,
Chris Cundy, *Stefano Ermon*, ICLR 2024
- 2024 **Privacy-Constrained Policies via Mutual Information Regularized Policy Gradients**,
Chris Cundy, *Rishi Desai*, *Stefano Ermon*, AISTATS 2024
- 2022 **LMPriors: Pre-Trained Language Models as Task-Specific Priors**,
*Kristy Choi**, **Chris Cundy***, *Sanjari Srivasta*, *Stefano Ermon*, First Workshop on Foundation Models for Decision Making, NeurIPS 2022
- 2021 **BCD Nets: Scalable Variational Approaches for Bayesian Causal Discovery**,
Chris Cundy, *Aditya Grover*, *Stefano Ermon*, NeurIPS 2021
- 2020 **Flexible Approximate Inference via Stratified Normalizing Flows**,
Chris Cundy, *Stefano Ermon*, UAI 2020
- 2018 **Parallelizing Linear Recurrent Neural Nets over Sequence Length**,
Eric Martin, **Chris Cundy**, ICLR 2018

Additional Publications

- 2026 **Scaling Trends for Lie Detector Oversight in Preference Learning**,
Oskar J. Hollinsworth, *Ann-Kathrin Dombrowski*, *Sam Adam-Day*, *Adam Gleave*, **Chris Cundy**, AI4Good Workshop, ICML 2026
- 2026 **The Safety Gap Toolkit: Evaluating Hidden Dangers of Open-Source Models**,
Ann-Kathrin Dombrowski, *Dillon Bowen*, *Adam Gleave*, **Chris Cundy**, LockLLM Workshop, NeurIPS 2025
- 2025 **Sharpe Ratio-Guided Active Learning for Preference Optimization in RLHF**,
Syrine Belakaria, *Joshua Kazdan*, *Charles Marx*, **Chris Cundy**, *Willie Neiswanger*, *Sanmi Koyejo*, *Barbara E Engelhardt*, *Stefano Ermon*, CoLM 2025
- 2024 **A physics-informed machine learning model for the prediction of drop breakup in two-phase flows**,
Chris Cundy, *Shahab Mirjalili*, *Charl  lie Laurent*, *Stefano Ermon*, *Gianluca Iaccarino*, *Ali Mani*, International Journal of Multiphase Flow
- 2023 **Neural Networks and the Chomsky Hierarchy**,
Gr  goire Del  tang, *Anian Ruoss*, *Jordi Grau-Moya*, *Tim Genewein*, *Li Kevin Wenliang*, *Elliot Catt*, **Chris Cundy**, *Marcus Hutter*, *Shane Legg*, *Joel Veness*, *Pedro A. Ortega*, ICLR 2023
- 2022 **Towards a foundation model for geospatial artificial intelligence**,
Gengchen Mai, **Chris Cundy**, *Kristy Choi*, *Yingjie Hu*, *Ni Lao*, *Stefano Ermon*, Proceedings of the 30th International Conference on Advances in Geographic Information Systems
- 2021 **IQ-Learn: Inverse soft-Q Learning for Imitation**,
Divyansh Garg, *Shuvam Chakraborty*, **Chris Cundy**, *Jiaming Song*, *Stefano Ermon*, NeurIPS 2021
- 2018 **Exploring Hierarchy-Aware Inverse Reinforcement Learning**,
Chris Cundy, *Daniel Filan*, Workshop on Goal Specifications for Reinforcement Learning, ICML 2018
- 2017 **Predicting Slow Judgment**,
Owain Evans, *Andreas Stuhlm  ller*, *Ryan Carey*, *Neal Jean*, *Andrew Schreiber*, *Girish Sastry*, **Chris Cundy**, Aligned Artificial Intelligence Workshop, NeurIPS 2017

Service

- 2025 **Participant, EU AI Act Code of Practice Working Groups 2 and 4**
Participated, as an independent expert, in working groups 2 and 4 for the development of the EU AI Act Code of Practice (CoP). I advocated, via oral presentation, for both pre- and post-mitigation model evaluations
- 2023 **Teaching Assistant–CS228 (Probabilistic Graphical Models)**, *Stanford University*

- 2022 **Head Teaching Assistant–CS228 (Probabilistic Graphical Models)**, *Stanford University*
Received award for excellence (awarded to top 5% of Teaching Assistants).
- 2023 **Project Supervisor**, *Supervised Project for Alignment Research (SPAR)*, Stanford AI Alignment
Supervised five undergraduates on a project finding scaling laws in prompt injections.
- 2021 **Project Supervisor**, *Undergraduate Research Program*, Stanford Existential Risk Initiative
Served as supervisor for an undergraduate project on forecasting AI progress.
- 2020– **Reviewer**
Reviewed for the following venues: UAI (2020,2022,2025,2026), ICML (2019,2020,2023,2025,2026), ICLR (2021-2026), NeurIPS (2021-2025), AAAI-(Safe and Robust AI track) (2023-2024), TMLR (2025,2026)

Relevant Awards

- March 2024 **Winner, OpenAI Preparedness Challenge**
- One of the top ten submissions for the OpenAI Preparedness Challenge, for submitting *the most unique, while still being probable, potentially catastrophic misuse of the [OpenAI API]*.
 - Developed proof-of-concept showing how GPT4-V, and speech-to-text with GPT4, could be used to parse vast amounts of unlabelled surveillance data, finding actionable insights for blackmail or insider trading.